

ASSIGNMENT NO. 3

PROGRAMMING FUNDAMENTALS

Question 1: Write a function power (a, b), to calculate the value of a raised to b.

Question 2: 5-digit positive integer is entered through the keyboard, write a function to calculate sum of digits of the 5-digit number: (1) Without using recursion (2) Using recursion

Question 3: Write a program in C language to obtain the numbers of a Fibonacci sequence UPTIL THE USER REQUIRES. In a Fibonacci sequence the sum of two successive terms gives the third term. Following are the first few terms of the Fibonacci sequence: **1 1 2 3 5 8 13 21 34 55 89...**

Question 4: What would be the output of the following programs:

(a)

```
main( )
{ int i = 0 ; i++ ; if ( i <= 5 )
  { printf ( "\nC adds wings to your thoughts" ) ;
    exit( ) ;
    main( ) ;
  }
}
```

(b)

```
main( )
{ static int i = 0 ; i++ ;
  if ( i <= 5 )
  { printf ( "\n%d", i ) ;
    main( ) ;
  }
  else exit( ) ;
}
```